

CYB0X-S: OPERATOR GUIDE & PRACTICAL WORKFLOW REFERENCE

High-Speed Offline Penetration Testing Worksheet · eJPTv2 / eCPPT Practical Companion

What is CYB0X-S?

CYB0X-S is a fast, keyboard-driven terminal worksheet and operational cockpit designed to eliminate exam cognitive overload. It provides an offline state machine for host discovery, port tracking, credential reuse, syntax cheatsheets, and question proofs. **It does not run autonomous exploits or rely on AI.** You retain 100% human control while CYB0X-S manages your operational memory.

INE eJPT / eCPPT Exam Compliance

- **100% Passive & Offline:** Runs entirely on localhost via local SQLite database.
- **Zero Autonomous Action:** Commands are copied to your clipboard; YOU execute them.
- **Zero Cloud / AI Dependencies:** No external API calls, leaks, or prohibited LLMs.
- **Permitted Personal Notes:** Functions strictly as a local worksheet and syntax lookup.

1. The Four Operational Stations (Switch with Keys 1, 2, 3, 4)

Station	Name & Purpose	What You See & Do Here	Hotkey
Station 1	Cockpit (Workbench)	Targets tree, open ports, service triage status, methodology checklist, scratchpad notes, and interactive command console.	[1]
Station 2	Playbook Browser	Browse full tactical playbooks (eJPT workflow, Web App OWASP, Active Directory, Pivoting, Linux & Windows PrivEsc) with instant copy.	[2]
Station 3	Credential Matrix	2D grid of discovered credentials (rows) × target services (cols). Press [Enter] to compile & copy spray commands; [Space] to cycle verification state.	[3]
Station 4	Exam Proofs & Loot	Track exam question answers (Q1–Q35 via :q or [a]), user/root flags, and Failure Logs . Run :export exam for offline Markdown submission dossier.	[4]

2. Station 1 Cockpit Architecture & Real-Time Workspace

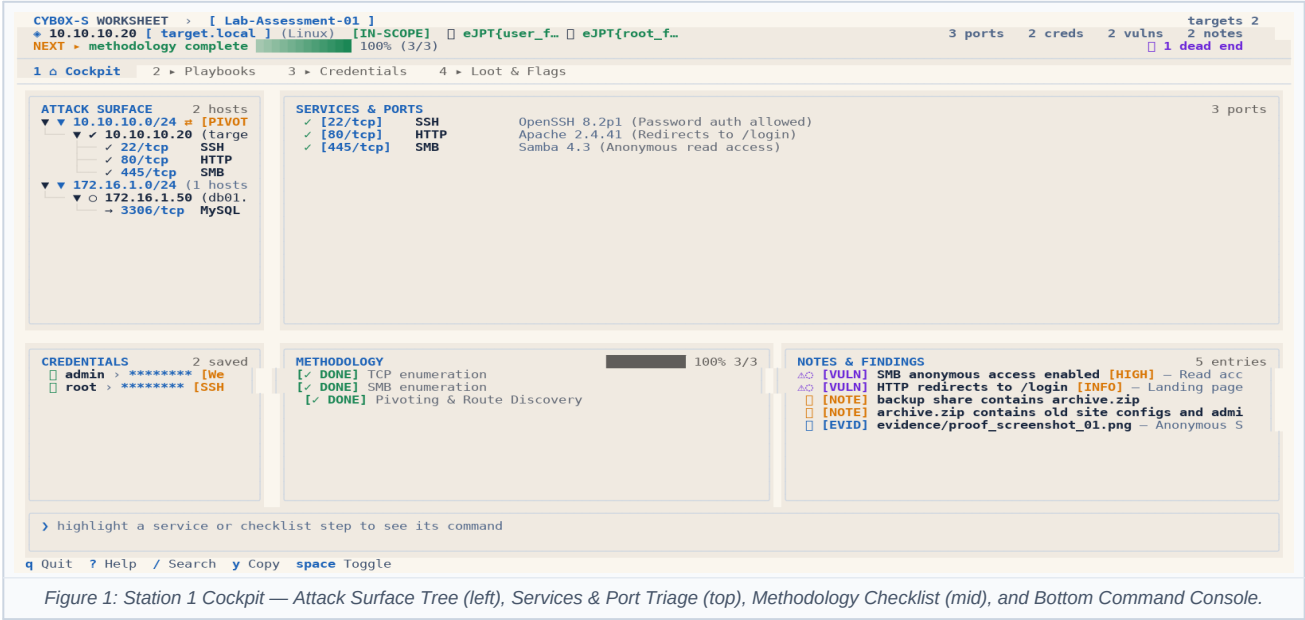


Figure 1: Station 1 Cockpit — Attack Surface Tree (left), Services & Port Triage (top), Methodology Checklist (mid), and Bottom Command Console.

3. Complete Keyboard Shortcuts Cheatsheet ("Muscle Memory Map")

CYBOX-S is 100% operational from the keyboard. Single-letter keys trigger immediate actions unless an input modal is active:

Key	Category	Action & Operational Behavior
1, 2, 3, 4	Station	Switch immediately to Station 1 (Cockpit), 2 (Playbooks), 3 (Creds Matrix), or 4 (Loot).
w	Navigation	Cycle Panels: Sequentially rotate focus across all panels in Cockpit without Tab.
h / l	Navigation	Column Jump: Press h for Sidebar (left); press l for Workbench (right).
j / k	Navigation	Vim Navigation: Move highlight down (j) or up (k) in the active list or tree.
[/]	Targeting	Target Switcher: Switch immediately to previous [or next] target machine.
b	Display	Sidebar Toggle: Collapse sidebar to give 100% width to Workbench (press b again to restore).
: (colon)	Console	Focus Command Bar: Instantly activates the bottom console input ready to type.
Esc	Console	Dismiss / Blur: Clears command input and returns focus directly to the active workbench.
Space	Triage / Reveal	• On Service: Cycle triage status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED]. • On Credential: Reveal/mask secret password. • In Station 3 Matrix: Cycle test state: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].
, / .	Recipe Carousel	Tool Carousel: When a service is highlighted, press . (next) or , (prev) to cycle alternative tool recipes in the console (e.g. feroxbuster → gobuster → nikto → curl).
Enter	Execute / Copy	• On Guidance / Service: Copies the previewed tool command to system clipboard. • In Station 3 Matrix: Compiles & copies ready-to-run spray command for credential & port.
y	Quick Copy	Copies selected entity's primary value (IP address, port, password, or checklist command).
z	Layout	Zoom: Maximize the focused panel to full-screen; press z again to restore normal layout.
T	Theme	Theme Picker: Open visual theme modal with 10 vibrant palettes (press d to set default).
/ or Ctrl+F	Search	Global Fuzzy Search: Search across targets, services, credentials, checklists, and notes.
r	Reference	Quick Reference Modal: Open offline practical cheatsheet for instant syntax lookup.
t / s / c / n	Fast Capture	Modal dialogs to quickly add Target (t), Service (s), Credential (c), or Note (n).
a	Exam Proof	In Station 4: Open Add Exam Proof modal to link question number (Q1–Q35) to proof evidence.



Figure 2: Practical Reference Cheatsheet Modal (Hotkey [r]) — Instant syntax lookup for Nmap, NetExec, Hydra, Chisel, and LinPEAS.

PRO-TIP: THE ZERO-MOUSE OPERATOR FLOW

1. Highlight any port in **SERVICES & PORTS** with j / k.
2. Watch the bottom console display the exact syntax with the target IP already substituted.
3. Press . to cycle alternative tools (e.g., feroxbuster → gobuster → nikto → curl).
4. Press Enter to copy the command directly to your terminal clipboard, switch to your tmux/terminal pane, and run it!
5. Press Space to mark the port [CHECKED] or [DEAD-END].

4. Bottom Console Syntax & Wordlist Accelerators

Console Command	Description & Operational Effect	Example Syntax
:w <alias>	Wordlist Accelerator: Instantly copies standard Kali / SecLists paths to clipboard.	:w rockyou :w common :w medium :w raft-d :w raft-f :w users
:t <ip> [opts]	Add target with optional host, OS, subnet, and pivot flag.	:t 10.10.10.5 host linux 10.10.10.0/24 pivot
:pivot / :subnet	Tag pivot machine / assign subnet CIDR for Station 1 tree grouping.	:pivot 10.10.10.5 172.16.1.0/24 :subnet 10.10.10.5 10.10.10.0/24
:s <port> <svc>	Record newly discovered port & service on active target.	:s 445/tcp smb :s 8080 http
:c <user:pass> [scope]	Record discovered credential with optional service scope.	:c admin:Secret123! SMB :c root:toor SSH
:q / :export	Pin exam question proof (Q1–Q35) or export markdown exam dossier.	:q 7 /var/www/wp-config.php :export exam :export report
:uflag / :rflag <hash>	Record user flag or root/admin flag into Station 4 Loot Ledger.	:uflag 7a9e... :rflag f04b...
:st <where> / :cl <clue>	Record stuck point (rabbit hole) and breakthrough clue.	:st stuck on wp-login :cl found backup.zip
:m <template>	Instantiate standard checklist template (replace or append).	:m ejpt :m web :m smb append
:theme <name>	Switch theme live (e.g. sugary, midnight, slate, caramel, cyber).	:theme sugary :theme midnight

5. Step-by-Step Practical Exam Workflow ("The Battle Plan")

PHASE 1: SCOPING & HOST DISCOVERY

1. Host Discovery: Identify alive target IPs (nmap -sn 10.10.10.0/24 or arp-scan).

2. Fast Add to CYBOX-S: Type :t 10.10.10.20 into console. Assign subnets: :subnet 10.10.10.20 10.10.10.0/24.

3. Scope & Routing: Tag dual-homed pivot machines with :pivot 10.10.10.20 172.16.1.0/24 for automatic guidance.

4. Import XML Option: Mass scans can be imported directly: cybox-s import /path/to/scan.xml.

PHASE 2: DETAILED PORT & SERVICE ENUMERATION

1. Full Port Sweep: nmap -sS -p- --min-rate 1000 <IP>, then version scan: nmap -sV -sC -p <ports> <IP>.

2. Record Open Services: Type :s 80 http, :s 445 smb, :s 3306 mysql into the console.

3. Inspect Tactical Guidance: Highlight each service in SERVICES & PORTS. Press . to cycle tool recipes (whatweb → feroxbuster → gobuster → nikto → curl). Press [Enter] to copy.

4. Service Triage with [Space]: Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED].

PHASE 3: WEB & LOW-HANGING FRUIT EXPLOITATION

1. Inspect Anonymous / Default Access: Check anonymous FTP (:s 21), null SMB shares (:s 445), and SNMP strings (:s 161).

2. Directory Fuzzing: Type :w common or :w medium to copy the SecLists directory path. Paste directly into feroxbuster or gobuster.

3. Cross-Filtering: Highlighting a service in SERVICES & PORTS auto-scrolls the **Methodology** checklist to that service's testing steps!

PHASE 4: CREDENTIAL SPRAYING & LATERAL MOVEMENT (STATION 3)

1. Record Every Credential: Whenever you find creds (web configs, DB dumps, code), press [c] or type :c user:pass scope.

2. Open Station 3 Matrix: Press [3] to switch to the 2D Credential Spray Matrix.

3. Instant Spray Generation: Highlight any cell and press [Enter] to compile & copy ready-to-run spray commands (netexec, evil-winrm).

4. Record Verification State: Press [Space] on the cell to cycle: [UNTESTED] → [VALID] → [PWN3D] → [INVALID]. State persists in SQLite!

PHASE 5: PRIVESC, PIVOTING & EVIDENCE CAPTURE (STATION 4)

1. Pivoting & Route Addition: Tag pivot hosts with :pivot to see dynamic proxy guidance in the console bar.

2. Privilege Escalation: Switch to Station 2 (press [2]) and search Linux SUID, sudo -l, Cron jobs, or Windows tokens / services.

3. Capture Question Evidence & Flags: Log question proofs with :q <num> <proof> (or press [a] in Station 4). Record flags with :uflag / :rflag.

4. Export Exam Submission Bundle: Run :export exam to write a clean Markdown report with all Q1–Q35 evidence, flags, and pivot paths!

THE 20-MINUTE ROTATION RULE (EXAM GOLDEN RULE)

If you have been analyzing a single service for more than 20 minutes without a new lead: **STOP**. Press [Space] on the service to mark it [DEFERRED]. Type :stuck <service description> to log where you are stuck, press] to rotate to another machine or port, and harvest low-hanging fruit elsewhere. 80% of exam failures occur because candidates spend 3 hours trapped down an unexploitable rabbit hole!

6. Station 3 Deep Dive: 2D Credential Spray Matrix

In multi-machine exams (like eJPTv2 / eCPPT), credential reuse across SSH, SMB, WinRM, and Web portals is the primary lateral movement technique. Station 3 prevents credential amnesia by organizing all discovered credentials (rows) against all discovered target services (columns) in an interactive 2D grid:

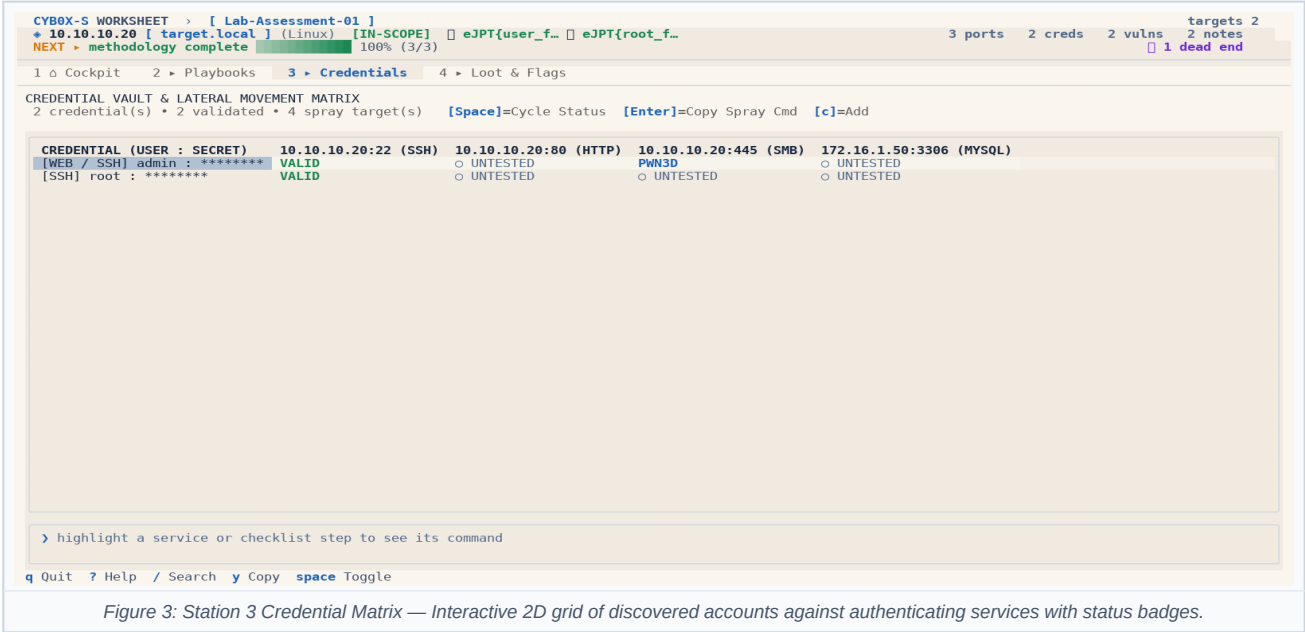


Figure 3: Station 3 Credential Matrix — Interactive 2D grid of discovered accounts against authenticating services with status badges.

Action / Key	Mechanism & Tactical Utility	Persistence & Exam Value
Arrow Keys / Vim	Navigate freely across cells in the 2D grid.	Instant visual overview of which accounts were tested where.
[Space] (Cycle State)	Cycles through: [UNTESTED] → [VALID] (Green) → [PWN3D] (Blue) → [INVALID] (Plum).	Persisted instantly to local SQLite database. Never guess if you tried a password.
[Enter] (Copy Command)	Automatically compiles and copies the exact command line to your clipboard (e.g. netexec smb 10.10.10.20 -u admin -p 'P@ssword123').	Zero typos under stress; supports SSH, SMB, WinRM, RDP, MySQL, and MSSQL.
[c] (Fast Add)	Opens modal to record newly discovered credentials (or type :c user:pass [scope]).	Automatically expands the matrix with a new row across all existing targets.

LATERAL MOVEMENT WORKFLOW SUMMARY:

- 1. **Harvest:** Extract credentials from web app configs (wp-config.php, .env), DB dumps, or SAM registry hive.
- 2. **Input:** Press [c] in CYBOX-S to add the username and password.
- 3. **Matrix Spray:** Press [3] to open the grid. Highlight the intersection of the new credential with port 445 (SMB) or 22 (SSH).
- 4. **Execute:** Press [Enter] to copy the pre-built NetExec/SSH command, paste and run in your terminal.
- 5. **Record:** If admin/pwn3d, press [Space] twice to set [PWN3D]. You now have your next pivot point!

7. Station 4 Deep Dive: Exam Question Proofs (Q1–Q35) & Loot Ledger

The eJPTv2 exam consists of 35 dynamic practical questions that require concrete evidence: hashes, specific directory names, database records, usernames, or service versions. Station 4 provides a dedicated ledger to link lab findings directly to question numbers, plus a Failure Log to manage stuck points:

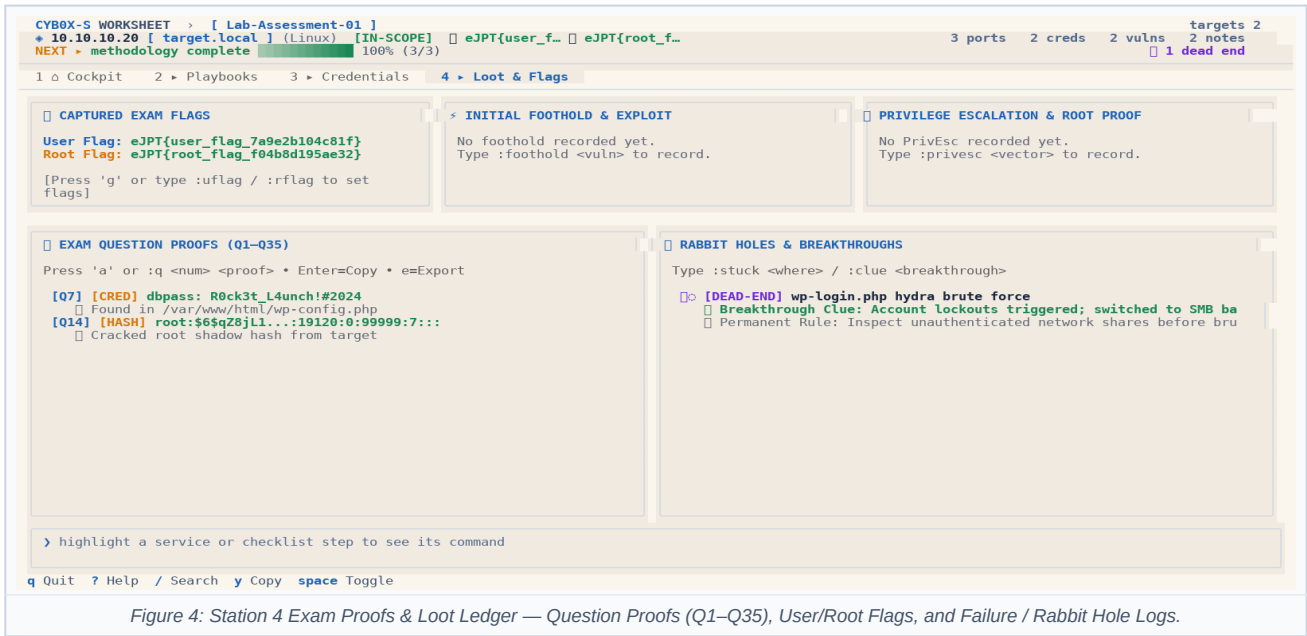


Figure 4: Station 4 Exam Proofs & Loot Ledger — Question Proofs (Q1–Q35), User/Root Flags, and Failure / Rabbit Hole Logs.

Module / Feature	Hotkeys & Console Syntax	Operational Purpose & Deliverable
Question Proofs (Q1–Q35)	Press [a] (modal) or :q <num> <proof> [notes]	Links evidence directly to an exam question number (e.g. :q 14 /etc/passwd root hash). Prevents having to re-exploit targets at exam end.
Loot & Flags Tracker	:uflag <hash> (user flag) :rflag <hash> (root flag)	Captures proof hashes with timestamps and associated target IP. Safely stored in local SQLite database.
Failure Log (Anti-Rabbit Hole)	:st <stuck_point> :cl <breakthrough_clue>	Records dead-ends and breakthroughs so you never repeat failed brute-force or injection attempts.
Markdown Submission Dossier	Type :export exam or :export report in console	Generates an instant offline Markdown report (exam_proofs.md) summarizing all answered questions, flags, and credentials.

THE OFFLINE SUBMISSION DOSSIER (:export exam):

At any time, or before finalizing your exam answers in the INE portal, run :export exam in the bottom console. CYBOX-S exports a complete, self-contained Markdown file containing: (1) Target Inventory & Subnet Map, (2) Solved Questions 1–35 with exact evidence strings, (3) Captured User & Root Flags, (4) Discovered Credentials & Verification States, and (5) Audit Trail of all tested services. Keep this document open in your text editor while answering the 35 exam questions to achieve 100% submission confidence.

8. Station 2 Playbooks & Built-In Methodology Templates

Station 2 serves as an offline tactical encyclopedia. You can search playbooks for specific attack chains, inspect commands, and copy snippets with a single keystroke:



Figure 5: Station 2 Playbook Browser — Tactical guides, command recipes, and methodology references with instant keyboard copy.

Template Alias	Category	Key Checklist Steps & Focus
:m ejpt	eJPT Practical	Scope recon → Host discovery → Full TCP scan → Service triage → Low-hanging fruit → Web fuzzing → Cred spray → Pivoting → PrivEsc.
:m web	OWASP Web App	Technology profiling (whatweb) → Directory fuzzing (feroxbuster) → Parameter fuzzing → SQLi → Auth bypass → LFI/RFI → File upload.
:m smb	SMB & Windows	Null session shares (smbclient/smbmap) → User enum (rpcclient) → Password policy → Anonymous signing audit → Vulnerability scan (MS17-010).
:m pivoting	Network Pivoting	Dual-homed adapter discovery (ip a) → Internal route inspect → Proxychains & Chisel server setup → Port forward (socat) → Subnet sweep.
:m privesc_linux	Linux PrivEsc	sudo -l commands → SUID/SGID binaries (GTFEBins) → Capabilities (getcap) → World-writable scripts & crontabs → Internal listening ports.
:m privesc_windows	Windows PrivEsc	whoami /priv (SeImpersonate) → Unquoted service paths → Modifiable service binaries → Stored credentials (cmdkey) → Registry backup.

EXAM START PROCEDURES:

1. Open terminal on your host or VM and launch: cyb0x-s

2. Pick your visual theme: press [T], select palette (e.g. Sugary, Midnight, Slate, or Cyber), press [d] to save as default.

3. Import initial Nmap scan: run cyb0x-s import /path/to/scan.xml or fast-add targets with :t <ip>.

4. **Offline Markdown Submission Bundle:** Run :export exam at any time to output a structured markdown report containing all Q1–Q35 evidence, flags, credentials, and pivot paths.

5. **Database Safety:** Stored in ~/.local/share/cyb0x-s/worksheets.db. 100% local, persistent, zero network calls, and fully compliant with INE exam rules.